RGCS Phryll v2 — Build Sheet

Design PHV2-CRY-DEMO-120 · crystal CRY-DEMO-120

Crystal profile

crystal ID	CRY-DEMO-120
length (mm)	120
top / base diameter (mm)	26 / 39
Eye z (mm)	62.5

Generated parts

custom cone shell, coil sleeve with grooves, base adapter, cap, LED holder, jack holder, locker — all generated from the crystal envelope; no stock mesh scaled

Print settings

wall (mm)	1.8
clearance (mm)	0.66
print tolerance (mm)	0.2
suggested material	PLA / PETG

Coil settings (crossed ±45° multi-start lattice)

wire	AWG28 (0.33 mm)
copper	enameled copper — clockwise at +45°
silver	silver or non-ferrous source-selected conductor — counter_clockwise at -45°
starts per coil	83
rise per turn (mm)	116.699
turns per strand across band	0.514
winding band (mm)	32.5 – 92.5
lattice crossing	X centered on the Eye plane
electrical contact	none permitted between coils

Wire spacing

clear gap, perpendicular (mm)	0.66
strand pitch, perpendicular (mm)	0.99
axial strand spacing (mm)	1.40007
groove depth (mm)	0.25

Coil-to-crystal standoff

nearest conductor (mm)	2.21
coil centerline (mm)	2.375

Eye alignment

Eye z (mm)	62.5
crossing plane z (mm)	62.5
alignment residual (mm)	0
tolerance (mm)	0.5
aligned	PASS

Crystal-bottom coupling (crystal bottom -> gap -> flat pickup -> annular ring; bottom stays open)

coupling mode	open
gap (mm)	2
pickup ring OD / ID (mm)	45 / 31
O-ring	none (open coupling)
bottom aperture	open — no solid plastic under the crystal base

Excitation paths (hardware first)

1	photonic / laser
2	magneto-acoustic / pulsed coils
3	mechanical / acoustic
4	electrical / coil
intention/focus-only	source-language only — not an implemented mode

Pulse metadata (source-language, recorded not validated)

drive	two coils pulsed alternately at 4096 Hz
listed starting voltage	20 V (source-reported; user limits govern)
crossing	copper/silver crossed, no contact

Assembly notes

Wind copper clockwise and silver counter-clockwise in their grooves; verify the crossing plane against the Eye mark before fixing; check no electrical contact between coils with a continuity meter.

Claim boundary

This build sheet is an engineering plan and reproducibility record. Source-language pulse notes are recorded, not validated. Predictions are model outputs. Measurements decide.

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